Topic 6.1 - Mole Ratio

1. What is needed to determine mole ratios?

2. Mole ratio is:

3. What number gives the number of moles of a substance involved in the reaction?

Find the mole ratios to the following:

1. Given this equation: $N_2 + 3\ H_2 \text{ ---> } 2\ NH_3$, write the following mole ratios:
   a) N_2 / H_2 _____
   b) N_2 / NH_3 _____
   c) H_2 / NH_3 _____

2. Given the following equation: $8\ H_2 + S_8 \text{ ---> } 8\ H_2S$, write the following mole ratios:
   a) H_2 / H_2S _____
   b) H_2 / S_8 _____
   c) H_2S / S_8 _____

3. Water can be produced from a reaction between hydrogen gas and oxygen gas.
   a) Write the balanced chemical equation:

   b) What are the mole ratios for: hydrogen to oxygen _____
      hydrogen to water _____
      oxygen to water _____
   c) How many moles of water can be produced from 10 moles of hydrogen.

4. Use this equation: $N_2 + 3\ H_2 \text{ ---> } 2\ NH_3$, for the following problems
   a) If 10 moles of NH_3 were produced, how many moles of N_2 would be required?

   b) If 3.00 moles of H_2 were used, how many moles of NH_3 would be made?

   c) If 0.600 moles of NH_3 were produced, how many moles of H_2 are required?

5. The reaction between hydrazine (N_2H_2) and hydrogen peroxide (H_2O_2) is used to fuel rockets. The products are nitrogen gas and water.
   a) Write a balanced chemical equation: _____

   b) If 37 moles of hydrazine are used, how many moles of nitrogen gas are produced?

   c) If 23 moles of water are produced, how many moles of hydrogen peroxide were used?

Topic 6.2 - Mass to Mass Stoichiometry

1. Use the equation $Fe + S_8 \rightarrow FeS$ to answer the following:
   a) Balance the equation: _____

   b) How many moles of FeS are produced when 2.4 g of S_8 react?

   c) How many grams of FeS are produced when 5.9 grams of Fe is used.

   d) If 11.9 grams of sulfur is used, how many grams of iron is also used?

2. Use the equation $K_3PO_4 + Al(NO_2)_3 \rightarrow KNO_2 + AlPO_4$ to answer the following:
   a) Balance the equation _____

   b) What is the mass of potassium nitrite that is produced when 4.6 moles of potassium phosphate is used?

   c) If 5.8 grams of aluminum phosphate are formed, how many grams of aluminum nitrite is used?

   d) How many grams of potassium nitrite is formed when 6.2 grams of potassium phosphate is used?

3. Use the equation $Fe_2O_3 + C \rightarrow Fe + CO_2$ to answer the following:
   a) Balance the equation _____

   b) How many grams of iron are produced if 39.64 grams of carbon are used?

   c) If 63.12 grams of iron (III) oxide are used, how many grams of carbon dioxide are produced?

   d) If 38.2 grams of carbon are used, how many grams of iron are produced?

4. Use the equation $CO + H_2 \rightarrow CH_3OH$ to answer the following:
   a) Balance the equation: _____

   b) If 5.7 grams of carbon monoxide are used, how many grams of hydrogen gas are also used?

   c) If .89 grams of hydrogen gas were used, how much CH_3OH was produced?

   d) If 3.61 grams of carbon monoxide were used, how many grams of CH_3OH were produced?

Topic 6.3 - More Stoichiometry Conversions (Liters/Atoms)

1. Knowing that hydrogen gas and oxygen combine to form water, how many liters of Hydrogen gas are needed to produce 3 moles of water?

1b. According to the same equation, How many grams of water will be produced by 30.2L of hydrogen?

1c. According to the same equation, How many grams of water will be produced by 30.2L of oxygen?

1d. When 120.3 grams of water are produced, how many liters of hydrogen must've been used?

2. Use the equation $CO + H_2 \rightarrow CH_3OH$ to answer the following:

a) Balance the equation: _____

b) If 15.0 liters of carbon monoxide are used, how many liters of hydrogen gas are also used?

c) If 12 liters of hydrogen gas were used, how many grams of CH_3OH would be produced?

d) If 3.00 liters of carbon monoxide were used, how many grams of CH_3OH were produced?

3. Use the equation $NaN_3(s) \rightarrow Na(l) + N_2(g)$ to answer the following:

a) Balance the equation _____

e) How many liters of Nitrogen are produced if 15.62 grams of sodium nitride are used?

f) If 200.0 L of Nitrogen are produced, how many grams of sodium nitride were used?

g) If 31.4 grams of sodium are produced, how many liters of nitrogen are also produced?

4. Chlorine gas will react with solid sodium to produce sodium chloride. Balance this reaction to answer the following:

a) Balance the equation _____

h) How many grams of sodium chloride are produced if 15.0 liters of chloride gas are used?

i) If 150.0 L of Chlorine are consumed, how many grams of sodium chloride will be produced?

j) If 38.0 grams of solid sodium are used, how many liters of chlorine are produced?

Topic 5.4 - Limiting and Excess Reactants

1) 1.5 moles of CH_4 react with 2.5 moles of Cl_2 according to the equation below.

$$___CH_4 + ___Cl_2 \rightarrow ___CH_2Cl_2 + ___HCl$$

a. Using a BCA table, calculate the number of moles of product produced and the number of moles of excess reactant remaining.

2) 15.0 grams of N_2 gas reacts with 4.50 grams of H_2 gas according to the equation below.

$$___N_2 + ___H_2 \rightarrow ___NH_3$$

a. Using a BCA table, calculate the mass of the product produced and the mass of the excess reactant remaining.

3) 17.5 liters of NH_3 gas reacts with 12.5 liters of O_2 gas at STP according to the equation below.

$$___NH_{3\,(g)} + ___O_{2\,(g)} \rightarrow ___NO_{(g)} + ___H_2O_{\,(g)}$$

a. Using a BCA table, calculate the mass of each product produced. Also calculate the liters of the excess reactant remaining.

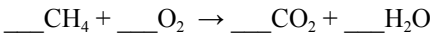
4) 7.50 grams of solid sodium metal reacts with 18.5 liters of Cl_2 gas at STP according to the equation below

$$___Na_{\,(s)} + ___Cl_{2\,(g)} \rightarrow NaCl_{\,(s)}$$

a. Using a BCA table, calculate the mass of NaCl produced and the mass of the excess reactant remaining.

5.5 - Percent Yield

1. A student performs the following reaction.

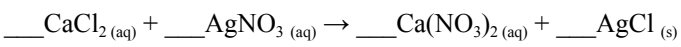


___CH_4 + ___O_2 → ___CO_2 + ___H_2O

   a. The student reacts 10.0 grams of oxygen gas with excess CH_4. Calculate the theoretical yield of carbon dioxide in the reaction.

   b. The student performs the lab and 6.50 grams of CO_2 is produced. Calculate the percent yield of the reaction.

2. A student performs the following reaction.

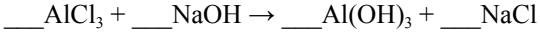


___$CaCl_{2\,(aq)}$ + ___$AgNO_{3\,(aq)}$ → ___$Ca(NO_3)_{2\,(aq)}$ + ___$AgCl_{\,(s)}$

   a. A student reacts 15.0 grams of calcium chloride and 10 grams of silver nitrate. Calculate the theoretical yield of silver chloride.

   b. The reaction is performed and 4.50 grams of silver chloride are made in the lab. Calculate the percent yield of the reaction.

3. A student performs the following reaction



___$AlCl_3$ + ___NaOH → ___$Al(OH)_3$ + ___NaCl

   a. 15.0 grams of aluminum chloride are reacted with 20.0 grams of sodium hydroxide. Calculate the theoretical yield of aluminum hydroxide in the reaction.

   b. If 11.3 grams of aluminum hydroxide are formed in the lab, what is the percent yield of this reaction?